Simultaneous formal extension of a first-order modular deformation and section (Proposition 4.5)

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Editorial note. Restore original source section/theorem counters at omitted §4.1.

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Editorial scope. Faithful material collection only; no assembly, compile, QA, mathematical repair or formal admission.

Allowed standard prerequisites (editorial).

- Formal Baker–Campbell–Hausdorff series. Original references [6] N. Bourbaki, Groupes et algèbres de Lie, Chapters II–III, Hermann, 1972; [15] E. B. Dynkin, Dokl. Akad. Nauk SSSR 57 (1947), 323–326.. Original source statement and finite-coefficient explanation at lines 1432–1460 retained; the current BCH cocycle cancellation Lemma 4.6 is retained in full, not outsourced.
- Group/sheaf cohomology identification and Stein vanishing. Original [16] S. Eilenberg and S. MacLane, Ann. of Math. (2) 46 (1945), 480–509; [29] T. Szamuely, Galois groups and fundamental groups, Cambridge University Press, vol.117, 2009; [4] H. Behnke and K. Stein, Math. Ann.120 (1947),430–461.. Standard external tools cited by authors in lines 479,784–791; retain the current application Lemma 2.8 and Theorems 3.12/4.3 in full. Higher coherent cohomology vanishing and nonabelian pointed-set exact sequences/inverse limits are used as the source uses them, without generating additional proofs.
- Bol's identity. The identity is used explicitly in author proof lines 1848–1855 and 1899–1905; the source gives no separately numbered bibliography item for Bol.. Standard weight -2 to weight 4 third-derivative intertwining, explicitly displayed in the retained equations. No invented citation or missing new core is substituted.
- Line-bundle deformations and modular-form context. Original [28] E. Sernesi, Deformations of algebraic schemes, Springer 2006, §§3.3.1,3.3.3 and Theorem3.3.11; [25] T. Miyake, Modular forms, Springer 1989.. Source definitions and stated classification retained. Brown regularized integral background is contextual only; no motivic cocycle construction is used by selected Proposition4.5.

Source reference. Original section 2.3: metric-decompositions/JEP338/sources/author-published.tex, source lines 850–931. Contextual comparison with an alternative analytic construction, not used in Proposition4.5.

Source reference. Original corollary 4.7: metric-decompositions/JEP338/sources/author-published.tex, source lines 1908–1941. Forward pointer in section introduction to consequence following Proposition4.5; not a premise.

Simultaneous formal extension of a deformation cocycle and section

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Original article and native mathematical source: CC BY 4.0, `https://creativecommons.org/licenses/by/4.0/`. This editable collection counts only Proposition 4.5, with its complete recursive proof and Lemma 4.6. Sections 1, 2.1–2.2, 3.1–3.3, the BCH setup from 3.4, and the relevant part of 4.2 are retained as author context and support, not additional claims. Bogo's examples, the remainder of 3.4, and the universal/motivic constructions in 4.1 and 4.3 are not dependencies of this counted proposition. The mathematical source is unchanged, including the original phrase "formal deformation cycle". This standard article wrapper, selection, numbering restoration and bibliography recovery from the publisher HTML are editorial preparation only; no independent mathematical certification, proof repair, formal verification or new assumptions are claimed. The full official original is supplied for context.

1 Preliminaries

We briefly recall the basic notions of slash actions on functions and differential operators, deformations of (the corresponding) line bundles, and multiple modular values.

1.1 Modular forms

Let $\mathbb{H}$ denote the complex upper half plane, and let $\overline{\mathbb{H}}$ be its closure. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{R})$ be a Fuchsian group, and let Γ act on the upper half plane by Möbius transformations

$$\gamma\tau := \frac{a\tau + b}{c\tau + d}, \quad \gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right).$$

For Γ acting freely and properly discontinuously on $\mathbb{H}$, the quotient space

$$Y_\Gamma := \Gamma\backslash\mathbb{H}$$

is a hyperbolic Riemann surface with fundamental group Γ [25].

Remark 1.1. For a less well behaved action, we should consider the orbifold quotient $\Gamma\backslash\backslash\mathbb{H}$ instead. In interest of simplicity, we shall not consider the orbifold situation, though all results should extend naturally.

In practice, we will be considering $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ acting freely and properly discontinuously on $\mathbb{H}$. For the sake of generality, we will refrain from making this restriction until Section 4. However, it should be noted that while a number of the results we reference are only explicitly proved for $\mathrm{SL}_2(\mathbb{Z})$, but hold for all subgroups.

For each $\gamma \in \Gamma$ define

$$j_\gamma(\tau) = c\tau + d, \quad \gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right).$$

We let L_k be the quotient of the trivial line bundle on $\mathbb{H}$ be the Γ-action

$$\gamma \cdot (\tau, z) = \big(\gamma\tau,\, j_\gamma^k(\tau) z\big).$$

Taking global sections of this line bundle, we obtain (weakly holomorphic) modular forms of weight k. In classical language this corresponds to functions satisfying the modular invariance property

$$(c\tau + d)^{-k} f\Big(\frac{a\tau + b}{c\tau + d}\Big) = f(\tau).$$

1.2 Functional multiple modular values

Brown introduces a class of iterated integrals associated to modular forms, defining periods of the (relative completion of the) fundamental group of modular curves, that he calls multiple modular values [9]. Brown's integrals are valued in certain polynomial representations of the modular group, but for our purposes it will be convenient to slightly modify this definition, to produce explicit components of Eichler-Shimura integrals.

Definition 1.2. Given cusp forms $h_1, \dots, h_d$ of weights $k_1, \dots, k_d$, we recursively define the functional multiple modular value by

$$\Lambda_\tau(h_1, \dots, h_d; n_1, \dots, n_d) := \int_\tau^{i\infty} h_1(z) z^{n_1} \, \Lambda_z(h_2, \dots, h_d; n_2, \dots, n_d) \mathrm{d}z$$

for integers n_i such that $0 \leq n_i \leq k_i - 2$ and $\tau \in \overline{\mathbb{H}}$. We define the empty integral by $\Lambda_\tau(-;-) := 1$.

Note that these are components of more classical Eichler-Shimura integrals, as in [9].

Remark 1.3. Brown defines functional multiple modular values for all modular forms and all exponents n_i, handling the divergent integrals arising via a regularisation procedure [9, 10, 24]. We do not give the full definition here, but take as given that the functional multiple modular value can be assigned a meaningful value.

Classical periods arise as special values of these multiple modular values [9, 10]. Note that the notation $\Lambda_\bullet$ introduced in the next definition strictly speaking overlaps with the one in Definition 1.2, however we expect that there will be no confusion between subscripts $\tau \in \mathbb{H}$ and $\gamma \in \Gamma$.

Definition 1.4. Given cusp forms $h_1, \dots, h_d$ of weight $k_1, \dots, k_d$, define the (classical) multiple modular value as the function

$$\Lambda_\bullet(h_1, \dots, h_d; n_1, \dots, n_d) : \Gamma \to \mathbb{C}$$

given by the integral

$$\Lambda_\gamma(h_1, \dots, h_d; n_1, \dots, n_d) := \Lambda_{\gamma^{-1}\infty}(h_1, \dots, h_d; n_1, \dots, n_d).$$

For example, the coefficients of the period polynomial associated to a weight-k cusp form h are precisely the values of the multiple modular value $\Lambda_\gamma(h; n)$ for $0 \leq n \leq k-2$, and are closely related to special values of the L-function associated to h.

1.3 The differential algebra as a Γ-module

The slash action on functions $\mathbb{H} \to \mathbb{C}$ has an analogue for differential operators. Letting $\mathcal{O}$ denote the algebra of holomorphic functions on $\mathbb{H}$, we consider the $\mathbb{C}$-vector space $\mathcal{O}[\partial_\tau]$ of finite order linear differential operators with holomorphic functions on $\mathbb{H}$ as coefficients. This may be equipped with a right action of Γ by conjugation with the weight k cocycle

$$\big(a\big\|_k \gamma\big)(\tau, \partial_\tau) := j_\gamma^{-k}(\tau) \, a\big(\gamma\tau, \partial_{\gamma\tau}\big) \, j_\gamma^k(\tau), \tag{1.1}$$

where $\partial_\tau = {}^d\!/_{d\tau}$ is the standard derivative. In this context, we record that

$$\partial_{\gamma\tau} = (c\tau + d)^2 \, \partial_\tau.$$

We write $\mathrm{ord}(a)$ for the order, as a differential operator, of an element $a \in \mathcal{O}[\partial_\tau]$.

Note that this action extends to an action of Γ on the ring of formal series

$$\mathcal{O}[\partial_\tau][\![\rho_1, \dots, \rho_n]\!]$$

by acting on the coefficients of a formal series.

1.4 Deformations of line bundles

We will construct deformations of modular forms by constructing sections of a deformation of the line bundle L_k. Following Sernesi [28] we define a deformation of a line bundle as follows. Note that we take a deformation of a line bundle as the simultaneous deformation of a line bundle and the base scheme ([28, §3.3.3]), not just the deformation of a line bundle over a fixed base scheme ([28, §3.3.1]), as such deformations are uninteresting in our case.

Definition 1.5. Let X be a scheme over a field K, and L a line bundle on X. A first order deformation of L consists of a pair $(\mathcal{X}, \mathcal{L})$ where

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \pi \\ \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}\big(K[\rho_1, \dots, \rho_n]/(\rho)^2\big) \end{array}$$

is a commutative diagram in which π is flat and surjective, $\mathcal{L}$ is a line bundle on $\mathcal{X}$ such that $L = \mathcal{L}|_X$, and (ρ) denotes the ideal generated by $\rho_1, \dots, \rho_n$.

Traditionally, a first order deformation has $n = 1$, but it will be convenient for us to consider this slightly wider definition.

Definition 1.6. Let X be a scheme over a field K, and L a line bundle on X. A formal deformation of L consists of a pair $(\mathcal{X}, \mathcal{L})$ where

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \pi \\ \mathrm{Spec}(K) & \longrightarrow & \mathrm{Specf}\big(K[\![\rho_1, \ldots, \rho_n]\!]\big) \end{array}$$

is a commutative diagram in which π is flat and surjective, and $\mathcal{L}$ is a line bundle on the formal scheme $\mathcal{X}$ such that $L = \mathcal{L}|_X$.

Definition 1.7. We say two first order (formal) deformations $(\mathcal{X}, \mathcal{L})$ and $(\widetilde{\mathcal{X}}, \widetilde{\mathcal{L}})$ of (X, L) are isomorphic if there exists an isomorphism of schemes $f : \mathcal{X} \cong \widetilde{\mathcal{X}}$ such that $\mathcal{L} \cong f^*\widetilde{\mathcal{L}}$, and f is compatible with the commutative diagrams.

When X is a nonsingular projective variety, the isomorphism classes of first order deformations are well understood. Denote by $\mathcal{O}_X$ the structure sheaf on X, and by T_X the tangent sheaf. Then, if L is a line bundle on X, the Chern class of L defines an extension

$$0 \to \mathcal{O}_X \to \mathcal{E}_L \to T_X \to 0,$$

where $\mathcal{E}_L$ is unique up to isomorphism, and referred to as the Atiyah extension associated to L.

Recall the Kodaira-Spencer theorem, which associates to a deformation of X a class in $\mathrm{H}^1(X, T_X)$. We refer to it as the Kodaira-Spencer class of the deformation. We then have the following result [28, Th. 3.3.11].

Theorem 1.8. *Let (X, L) be a pair consisting of a nonsingular projective algebraic variety X and a line bundle L on X. Then there is a canonical bijection between the space of isomorphism classes of first order deformations of (X, L) with $n = 1$ and the sheaf cohomology $\mathrm{H}^1(X, \mathcal{E}_L)$. Furthermore, the projection to $\mathrm{H}^1(X, T_X)$ is precisely the Kodaira-Spencer class of the deformation of X.*

For our more general definition of a first order deformation, an analogous result can be shown identically, with first order deformations being encoded by n-tuples of elements of $\mathrm{H}^1(X, T_X)$.

2 First order deformations and analytic deformations

The goal of this section is to give a rigorous definition of first order deformations of modular forms in Section 2.2, and then connect it to Bogo's results in Section 2.3. We require a specific differential Lie algebra that we examine and justify in Lemma 2.2 and the following remark. It leads naturally to deformation cocycles in Definition 2.4, which in turn come with deformation sections as specified in Definition 2.5. Deformations of modular forms then arise from the these deformation sections. A classification of first order deformation cocycles is possible and revisited in Proposition 2.9.

2.1 Cocycles associated with deformations of line bundles

For each weight k, we first define a Γ-submodule of $\mathcal{O}[\partial_\tau]$ by

$$\mathfrak{o}_k := \mathcal{O}\partial_\tau \oplus \mathcal{O}, \tag{2.1}$$

and equip it with the weight k action 1.1. Explicitly:

$$(a(\tau)\partial_\tau + b(\tau))\,\big\|_k\gamma = j_\gamma(\tau)^2 a(\gamma\tau)\partial_\tau + k j'_\gamma(\tau) j_\gamma(\tau) a(\gamma\tau) + b(\gamma\tau)$$

In the ensuing discussion, the first direct summand matches the tangent bundle in the Atiyah bundle, and the second one corresponds to $\mathcal{O}_X$. That is, by Theorem 1.8, the first term will correspond to deformations of the base, while the second one corresponds to deformations of the line bundle.

At times, we will use Fraktur letters, say, $\mathfrak{j}^{(k)}(\tau, \partial_\tau)$ for the elements of $\mathfrak{o}_k$ in order to emphasise that they are differential operators.

A first order deformation of the Riemann surface Y_Γ and the line bundle L_k as in Definition 1.5 is determined, up to isomorphism, by a n-tuple of elements of $\mathrm{H}^1(Y_\Gamma, \mathcal{E}_{L_k})$. The pullback of $\mathcal{E}_{L_k}$ to $\mathbb{H}$ splits as direct sum $T_\mathbb{H} \oplus \mathcal{O}$, corresponding to the two direct summands of $\mathfrak{o}_k$. Identifying the sheaf cohomology of Y_Γ with the group cohomology of Γ via this pullback [16, 29], we easily find that

$$\mathrm{H}^1(Y_\Gamma, \mathcal{E}_{L_k}) \cong \mathrm{H}^1(\Gamma, \mathfrak{o}_k). \tag{2.2}$$

Explicitly, we have the following:

Proposition 2.1. *A first order deformation of the Riemann surface Y_Γ and the line bundle L_k determines an element of*

$$\mathrm{H}^1(\Gamma, \mathfrak{o}_k[\rho_1, \dots, \rho_n]/(\rho)^2) \cong \left(\mathrm{H}^1(\Gamma, \mathfrak{o}_k)\right)^n$$

Proof. If we have a first order deformation $(\mathcal{Y}_\Gamma, \mathcal{L}_k)$ of (Y_Γ, L_k) we see that by pulling $\mathcal{L}_k$ back to a line bundle on $\mathbb{H}$, we must have that $\mathcal{L}_k$ is a quotient of the trivial line bundle. We see that the sections of $\mathcal{L}_k$ are can be identified with functions

$$f : \mathbb{H} \to \mathbb{C}[\rho_1, \dots, \rho_n]/(\rho)^2$$

such that, suppressing dependency on k,

$$f(\gamma\tau) = \mathrm{j}_\gamma^{(k)}(\tau, \partial_\tau)\, f(\tau)$$

for some linear differential operator

$$\mathrm{j}_\gamma^{(k)}(\tau, \partial_\tau) : \mathcal{O} \to \mathcal{O}[\rho_1, \dots, \rho_n]/(\rho)^2 \quad \text{with } \mathrm{j}_\gamma^{(k)}(\tau, \partial_\tau) \equiv j_\gamma^k(\tau) \pmod{(\rho)}$$

that satisfies

$$\mathrm{j}_{\gamma\delta}^{(k)}(\tau, \partial_\tau) = \mathrm{j}_\gamma^{(k)}(\delta\tau, \partial_{\delta\tau})\, \mathrm{j}_\delta^{(k)}(\tau, \partial_\tau) \quad \text{for all } \gamma, \delta \in \Gamma.$$

Writing

$$\mathrm{j}_\gamma^{(k)}(\tau, \partial_\tau) = j_\gamma^k(\tau) \left(1 + \sum_{i=1}^{n} \mathrm{j}_{\gamma,i}^{(k)}(\tau, \partial_\tau)\rho_i\right), \tag{2.3}$$

the above compatibility condition reduces using (1.1) to

$$\mathrm{j}_{\gamma\delta,i}^{(k)} = \mathrm{j}_{\gamma,i}^{(k)}\big\|_k \delta + \mathrm{j}_{\delta,i}^{(k)} \quad \text{for all } \gamma, \delta \in \Gamma. \tag{2.4}$$

This is precisely to say that each $\mathrm{j}_{\bullet,i}^{(k)}$ defines a 1-cocycle valued in $\mathfrak{o}_k$ and hence an element of $\mathrm{H}^1(\Gamma, \mathfrak{o}_k)$. □

However, the deformation theory of modular curves is fairly trivial, as this cohomology group vanishes for $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ (see Lemma 2.8), so we don't obtain much of interest. As such, we will restrict to deformations corresponding to cocycles that are algebraic in τ, that is to say to polynomial coefficients. The next lemma provides a suitable coefficient module.

Lemma 2.2. *Denote by $\mathbb{C}[\tau]_{\le 2}$ the space of polynomials in τ of degree at most two. Then the vector subspace*

$$\mathfrak{d}_k := \left\{2p(\tau)\partial_\tau + kp'(\tau) : p \in \mathbb{C}[\tau]_{\le 2}\right\} \tag{2.5}$$

of $\mathfrak{o}_k$ is a Γ-submodule.

Remark 2.3. This submodule is essentially the unique Γ-submodule spanned by polynomials. The coefficient of ∂_τ can be at most quadratic, and this uniquely determines the polynomial term up to addition of a constant. Our particular choice is motivated by the following observation.

Note that given a holomorphic modular form h the map

$$f \longmapsto 2\tilde{h}f' + k\tilde{h}'f$$

where $\tilde{h}$ is defined by 2.9, acts as a derivation on the (weight graded) ring of modular forms $M(\Gamma)$. It is for this reason that $\mathfrak{d}_k$ is defined as it is. Had we defined it naively as the largest submodule of $\mathfrak{o}_k$ contained in $\mathbb{C}[\tau]\partial_\tau \oplus \mathbb{C}[\tau]$, there would exist 1-cocycles that did not correspond to derivations, given by an element of $\mathrm{Z}^1(\Gamma, \mathfrak{d}_k)$ plus a homomorphism $\Gamma \to \mathbb{C}$. Deformations associated with the latter are studied in the theory of higher order modular forms [19, 12].

Proof. This is a direct computation. Let $\gamma \in \Gamma$ be given by the matrix $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Then

$$\begin{aligned}\big(2p(\tau)\partial_\tau + kp'(\tau)\big)\big\|_k\gamma &= 2j_\gamma^{-k}(\tau)p(\gamma\tau)\partial_{\gamma\tau}j_\gamma^k(\tau) + kp'(\gamma\tau)\\ &= 2j_\gamma^2(\tau)p(\gamma\tau)\partial_\tau + 2kj_\gamma'(\tau)j_\gamma(\tau)p(\gamma\tau) + kp'(\gamma\tau).\end{aligned}$$

Suppose $p(\tau) = p_2\tau^2 + p_1\tau + p_0$. Then the coefficient of ∂_τ is given by

$$2\big(a^2p_2 + acp_1 + c^2p_0\big)\tau^2 + 2\big(2abp_2 + adp_1 + bcp_1 + 2cdp_0\big)\tau + 2\big(b^2p_2 + bdp_1 + d^2p_0\big),$$

while the purely polynomial term reduces, using $ad - bc = 1$, to

$$2k\big(a^2p_2 + acp_1 + c^2p_0\big)\tau + k\big(2abp_2 + adp_1 + bcp_1 + 2cdp_0\big),$$

which easily seen to be an element of $\mathfrak{d}_k$. □

For later purpose, we record that we may equip both $\mathfrak{o}_k$ and $\mathfrak{d}_k$ with the structure of a Lie algebra as follows. Suppose

$$a(\tau) = f(\tau)\partial_\tau + g(\tau) \quad \text{and} \quad b(\tau) = r(\tau)\partial_\tau + s(\tau)$$

are elements of $\mathfrak{o}_k$. Then define

$$[a,b](\tau) := \big(f(\tau)r'(\tau) - r(\tau)f'(\tau)\big)\,\partial_\tau + f(\tau)s'(\tau) - r(\tau)g'(\tau) \tag{2.6}$$

as the commutator of a and b as differential operators. It is easy to check that this restricts to a Lie bracket on $\mathfrak{d}_k$, as

$$\big[2p\partial_\tau + kp', 2q\partial_\tau + kq'\big] = 4\big(pq' - p'q\big)\,\partial_\tau + 2k\big(pq'' - p''q\big)$$

and $pq' - p'q$ is a quadratic polynomial. Furthermore, direct computation shows that

$$\big[a\big\|_k\gamma, b\big\|_k\gamma\big] = [a,b]\big\|_k\gamma. \tag{2.7}$$

2.2 First order deformations of modular forms

We will now see that first order deformations of modular forms arise from the action of deformation sections, which are 0-cycles taking values in differential operators with holomorphic coefficients.

To identify meaningful 0-cycles, we choose those that trivialise a 1-cocycle taking values in differential operators with polynomial coefficients. This justifies our first Definition 2.4.

We emphasise how this approach parallels the relation between period polynomials and Eichler integrals. The period polynomial p_f associated with a weight-k cusp form f for, say, $\mathrm{SL}_2(\mathbb{Z})$ can be identified with a suitable 1-cocycle of $\mathrm{SL}_2(\mathbb{Z})$. In this language, the Eichler integral

$$\mathcal{E}_f := \int_0^{i\infty} f(z)(\tau - z)^{k-2}\,dz$$

associated with f are naturally 0-cocycles that trivialise them, as reflected by the coboundary equation

$$\mathcal{E}_f\big|_{2-k}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) - \mathcal{E}_f = p_f(\tau).$$

Definition 2.4. A first order algebraic deformation cocycle of weight k is an element

$$a_\bullet \in \mathrm{Z}^1\big(\Gamma,\, \mathfrak{d}_k[\rho_1,\dots,\rho_n]/(\rho)^2\big) \quad \text{with } a_\bullet(\tau,\partial_\tau) \equiv 0 \pmod{(\rho)}.$$

We identify it with a collection of n maps

$$a_{\bullet,i} : \Gamma \to \mathfrak{d}_k \quad \text{such that} \quad a_{\gamma\delta,i} = a_{\gamma,i}\big\|_k\delta + a_{\delta,i} \quad \text{for all } \gamma,\delta \in \Gamma$$

via the equation $a_\bullet = \sum_{i=1}^n a_{\bullet,i}\rho_i$.

We say two deformation cocycles $a_\bullet$, $\tilde{a}_\bullet$ are equivalent if they define the same class in the group cohomology $\mathrm{H}^1(\Gamma, \mathfrak{d}_k[\rho_1,\dots,\rho_n]/(\rho)^2)$. This is to say,

$$\tilde{a}_{\gamma,i} = a_{\gamma,i} + c_i\big\|_k\gamma - c_i \qquad \text{for some } c_i \in \mathfrak{d}_k \quad \text{and for all } i = 1,\dots,n.$$

We call the image of a first order algebraic deformation cocycle in cohomology a first order algebraic deformation class.

We will suppress the emphasis on algebraic and refer to cocycles in Definition 2.4 as first order deformation cocycles. In Proposition 3.5 we show that equivalent deformation cocycles define isomorphic deformations of L_k.

Next we look at the sections of the deformed line bundle, which we will call deformed modular forms.

Definition 2.5. A first order deformation section of weight k and deformation cocycle $a_\bullet$ is a trivialisation $b(\tau, \partial_\tau)$ under the coboundary map

$$\begin{aligned} \mathrm{C}^0\big(\Gamma,\, \mathfrak{o}_k[\rho_1,\dots,\rho_n]/(\rho)^2\big) \to \mathrm{Z}^1\big(\Gamma,\, \mathfrak{o}_k[\rho_1,\dots,\rho_n]/(\rho)^2\big) \supseteq \mathrm{Z}^1\big(\Gamma,\, \mathfrak{d}_k[\rho_1,\dots,\rho_n]/(\rho)^2\big) \\ b \longmapsto a_\bullet. \end{aligned}$$

We identify it via the equation $b = \sum_{i=1}^n b_i\rho_i$ with a collection of n elements

$$b_i \in \mathfrak{o}_k \quad \text{such that} \quad b_i\big|_k\gamma - b_i = a_{\gamma,i} \qquad \text{for all } i = 1,\dots,n \text{ and all } \gamma \in \Gamma.$$

Remark 2.6. It is worth emphasising that the b_i of Definition 2.5 are elements of $\mathfrak{o}_k$, not necessarily of $\mathfrak{d}_k$. Indeed, given a deformation cocycle

$$a_\bullet \in \mathrm{Z}^1(\Gamma, \mathfrak{d}_k)$$

the existence of a deformation section b of type $a_\bullet$ is equivalent to $a_\bullet$ being equivalent to the trivial deformation over $\mathfrak{o}_k$. That is to say that $a_\bullet$ is a 1-coboundary in $\mathrm{Z}^1(\Gamma, \mathfrak{o}_k)$. In particular, a necessary and sufficient condition for the existence of b is that the cohomology class of $a_\bullet$ vanishes under the natural map

$$\mathrm{H}^1(\Gamma, \mathfrak{d}_k) \to \mathrm{H}^1(\Gamma, \mathfrak{o}_k).$$

A deformation section b of weight k and cocycle $a_\bullet$ provides us with section of the deformed line bundle given by

$$\Big(\exp\big(b(\tau,\partial_\tau)\big)(f)\Big)(\gamma\tau) = \big(1 + b(\tau,\partial_\tau)\big)(f)(\gamma\tau) \tag{2.8}$$

for any modular form f via the action of elements of $\mathfrak{o}_k$ as differential operators. Note that this implies that, for a modular form f of weight k, the deformed modular form transforms as

$$\big(1 + b(\tau,\partial_\tau)\big)(f)(\gamma\tau) = j_\gamma^k(\tau)\Big(1 + \sum_{i=1}^n a_{\gamma,i}\Big)\,\big(1 + b(\tau,\partial_\tau)f\big)(\tau).$$

In particular, the $a_{\gamma,i}$ on the right hand side recover the cocycles $\mathfrak{j}_{\gamma,i}^{(k)}$ in (2.3), and b indeed provides a deformation of L_k of class $a_\bullet$.

To study deformation cocycles and deformation sections in weight k, it actually suffices to work in weight 0, as we have the following isomorphisms of Lie algebras.

Proposition 2.7. *There exists an isomorphism*

$$\begin{aligned} \phi_k : \mathfrak{o}_0 &\to \mathfrak{o}_k, \\ p\partial_\tau + q &\longmapsto p\partial_\tau + \tfrac{k}{2}p' + q, \end{aligned}$$

of Lie algebras for every k, compatible with the Γ-action. Furthermore, ϕ_k restricts to an isomorphism

$$\phi_k\big|_{\mathfrak{d}_0} : \mathfrak{d}_0^\rho \cong \mathfrak{d}_k.$$

Proof. We claim that the ρ-linear map induced by

$$\phi_k : \mathfrak{o}_0 \to \mathfrak{o}_k, \quad p\partial_\tau + q \longmapsto p\partial_\tau + \tfrac{k}{2}p' + q$$

is an isomorphism of Lie algebras with Γ-action. That it is an isomorphism of vector spaces is clear, as is the restriction to $\mathfrak{d}_0 \to \mathfrak{d}_k$. As such, it suffices to show that it is a homomorphism.

We first show that ϕ_k is compatible with the Γ-action. Note that

$$\begin{aligned} \phi_k\big((a\partial_\tau + b)\big|_0\gamma\big) &= \phi_k\big(j_\gamma^2(\tau)a(\gamma\tau)\partial_\tau + b(\gamma\tau)\big) \\ &= j_\gamma^2(\tau)a(\gamma\tau)\partial_\tau + kj_\gamma'(\tau)j_\gamma(\tau)a(\gamma\tau) + \tfrac{k}{2}a'(\gamma\tau) + b(\gamma\tau) \\ &= j_\gamma^{-k}(\tau)a(\gamma\tau)\partial_{\gamma\tau}j_\gamma^k(\tau) + \tfrac{k}{2}a'(\gamma\tau) + b(\gamma\tau) \\ &= \phi_k\big(a\partial_\tau + b\big)\big|_k\gamma. \end{aligned}$$

We then check that

$$\begin{aligned}\big[\phi_k(p\partial_\tau + r),\, \phi_k(q\partial_\tau + s)\big] &= \big[p\partial_\tau + \tfrac{k}{2}p' + r,\, q\partial_\tau + \tfrac{k}{2}q' + s\big] \\ &= \big[p\partial_\tau,\, q\partial_\tau\big] + \tfrac{k}{2}\big(pq'' - p''q\big) + ps' - qr' \\ &= \big(pq' - p'q\big)\partial_\tau + \tfrac{k}{2}\big(pq' - p'q\big)' + ps' - qr' \\ &= \phi_k\big(\big[p\partial_\tau + r,\, q\partial_\tau + s\big]\big).\end{aligned}$$

Thus ϕ_k is a Lie algebra isomorphism compatible with the Γ-actions. □

The existence of a deformation section is guaranteed by the following.

Lemma 2.8. *The first homology group* $\mathrm{H}^1(\Gamma, \mathfrak{o}_k) = 0$.

Proof. By Proposition 2.7, it suffices to show this for $k = 0$. Let Ω be the sheaf of Kähler differentials on Y_Γ. Via the standard identification of group cohomology with the cohomology of the modular curve with coefficients in a local system [16], we have that

$$\mathrm{H}^1\big(\Gamma, \mathfrak{o}_0\big) \cong \mathrm{H}^1\big(Y_\Gamma, \Omega^\vee\big) \cong 0,$$

as every Y_Γ is a Stein manifold [4] and hence has vanishing higher cohomology groups for sheaf cohomology valued in a line bundle. □

As it turns out, the classification of first order algebraic deformations, and choices of deformation section, are well known.

Proposition 2.9. *First order algebraic deformations are in bijection with n-tuples of elements of* $\mathrm{M}_4(\Gamma) \oplus \overline{\mathrm{S}_4(\Gamma)}$*, given by a pair of a weight* 4 *modular form and (the complex conjugate of) a weight 4 cusp form.*

Proof. As every element of $\mathfrak{d}_k$ is uniquely determined by the coefficient of ∂_τ, we have an isomorphism (Proposition 2.7)

$$\mathrm{H}^1(\Gamma, \mathfrak{d}_k) \cong \mathrm{H}^1\big(\Gamma, \mathbb{C}[\tau]_{\leq 2}\big),$$

where we equip $\mathbb{C}[\tau]_{\leq 2}$ with the Γ-action

$$p(\tau)\big\|\gamma := j_\gamma(\tau)^2\, p(\gamma\tau),$$

that is $\mathrm{Sym}^2(V)$ of the standard representation on a two dimensional vector space.

However, by the Eichler-Shimura isomorphism, we have

$$\mathrm{H}^1(\Gamma, \mathbb{C}[\tau]_{\leq 2})\mathrm{M}_4(\Gamma) \oplus \overline{\mathrm{S}_4(\Gamma)}.$$ □

The polynomials classifying first order deformations are called *period polynomials*. For example, via the Eichler-Shimura isomorphism, the deformation cocycle associated to a weight 4 cusp form is given by:

$$j_\gamma(\tau)^2\, \tilde{h}(\gamma\tau) - \tilde{h},$$

where

$$\tilde{h}(\tau) := \int_\tau^{i\infty} h(z)(\tau - z)^2\, \mathrm{d}z \tag{2.9}$$

is the holomorphic Eichler integral of the cusp form h. More generally, a period polynomial will be obtained by complex conjugation of coefficients, or a regularised Eichler integral of an Eisenstein series. In order to keep our computations concrete, we will only work with examples of first order deformations of the form

$$(bf)(\tau) = f(\tau) + \sum_{i=1}^{n} \big(2\tilde{h}_i(\tau)f'(\tau) + k\tilde{h}_i'(\tau)f(\tau)\big)\, \rho_i$$

for an n-tuple of cusp forms $(h_1, \ldots, h_n)$.

Remark 2.10. Note that, in the case where Γ admits a weight 1 modular form, this is precisely the first order deformation constructed via uniformising differential equations in [5] discussed in the following section.

3 Formal deformations

In this section, we pursue two goals. First, we establish the analogue of Section 2.2 on first order deformations of modular forms in the context of formal deformations. This requires us to set up the analogues $\mathbb{O}_k$ and $\mathbb{D}_k$ of the Lie algebras $\mathfrak{o}_k$ and $\mathfrak{d}_k$. They will serve in Section 3.2 as the value modules of formal deformation cocycles and formal deformation sections.

Second, we relate the formal deformations to first order ones. Every formal deformation yields a first order deformation by truncation. In the subject of deformations, however, not every first order deformation affords an extension to a formal one, and this is an important part of the theory. We show in Section 3.3 that in our setting every first order deformation extends. We give an existence argument leveraging vanishing of a suitable second cohomology group. It is not explicit, and therefore in Section 3.4 we illustrate how the extension to second order can be calculated.

3.1 Cocycles associated with deformations of line bundles

Suppose that we have constructed a formal deformation $(\mathcal{Y}_\Gamma, \mathcal{L}_k)$ of (Y_Γ, L_k) as in Definition 1.6. In analogy with first order deformations, by pulling $\mathcal{L}_k$ back to a line bundle on $\mathbb{H}$, we must have that $\mathcal{L}_k$ is the quotient of the trivial line bundle, and as such the sections of $\mathcal{L}_k$ can be identified with functions

$$f : \mathbb{H} \to \mathbb{C}[\![\rho_1, \dots, \rho_n]\!]$$

such that

$$f(\gamma\tau) = \mathcal{J}_\gamma^{(k)}(\tau)\, f(\tau)$$

for some linear differential operator

$$\mathcal{J}_\gamma^{(k)}(\tau, \partial_\tau) : \mathcal{O} \to \mathcal{O}[\![\rho_1, \dots, \rho_n]\!] \quad \text{with } \mathcal{J}_\gamma^{(k)}(\tau, \partial_\tau) \equiv j_\gamma^k(\tau) \pmod{(\rho)}$$

that satisfies

$$\mathcal{J}_{\gamma\delta}^{(k)}(\tau, \partial_\tau) = \mathcal{J}_\gamma^{(k)}(\delta\tau, \partial_{\delta\tau})\, \mathcal{J}_\delta^{(k)}(\tau).$$

Note that as in the case of first order deformations, we suppress the dependency on k.

Proceeding parallel to the case of first order deformations, we restrict our attention to $\mathcal{J}_\gamma^{(k)}(\tau, \partial_\tau)$ that we can write as

$$\mathcal{J}_\gamma^{(k)}(\tau, \partial_\tau) = j_\gamma^k(\tau)\, A_\gamma(\tau, \partial_\tau)$$

for some linear differential operator

$$A_\gamma(\tau, \partial_\tau) : \mathcal{O} \to \mathcal{O}[\![\rho_1, \dots, \rho_n]\!] \quad \text{with } A_\gamma \equiv 1 \pmod{(\rho)}.$$

The above conditions on $\mathcal{J}_\gamma^{(k)}(\tau)$ lead to the following constraints on A_γ:

$$A_{\gamma\delta}(\tau, \partial_\tau) = \big(j_\delta^{-k}(\tau) A_\gamma(\delta\tau, \partial_{\delta\tau}) j_\delta^k(\tau)\big) \cdot A_\delta(\tau, \partial_\tau) \quad \text{for all } \gamma, \delta \in \Gamma, \tag{3.1}$$

and, as already stated, $A_\gamma \equiv 1$ modulo $(\rho) = (\rho_1, \dots, \rho_n)$.

The first of these tells us that A_γ defines a non-abelian 1-cocycle in a certain Γ-module, while the second tells us that A_γ has a multiplicative inverse as a formal power series with coefficients in linear maps. Thus, we find that formal deformations are encoded in certain non-abelian 1-cocycles. However, the space of all such linear maps is too large for our purposes. We wish to restrict further to sufficiently "nice" spaces, so that we recover the theory of first order deformations modulo the ideal $(\rho)^2 = (\rho_1, \dots, \rho_n)^2$.

To obtain such spaces, we observe that we can equip both

$$\mathfrak{o}_k^\rho := (\rho)\, \mathfrak{o}_k[\![\rho_1, \dots, \rho_n]\!] \quad \text{and} \quad \mathfrak{d}_k^\rho := (\rho)\, \mathfrak{d}_k[\![\rho_1, \dots, \rho_n]\!] \tag{3.2}$$

with the structure of pro-nilpotent Lie algebras with Γ-action via the commutator

$$[a, b](\tau, \partial_\tau, \rho) := a(\tau, \partial_\tau, \rho) b(\tau, \partial_\tau, \rho) - b(\tau, \partial_\tau, \rho) a(\tau, \partial_\tau, \rho).$$

Note that $\mathfrak{d}_k^\rho$ is a Lie subalgebra of $\mathfrak{o}_k^\rho$. Hence, we can define their exponential groups

$$\mathbb{O}_k := \exp\big(\mathfrak{o}_k^\rho\big) \subset \mathcal{O}[\partial_\tau][\![\rho_1, \dots, \rho_n]\!]^* \text{ and } \mathbb{D}_k := \exp\big(\mathfrak{d}_k^\rho\big) \subset \mathbb{C}[\tau, \partial_\tau][\![\rho_1, \dots, \rho_n]\!]^* \tag{3.3}$$

as subgroups of the multiplicative group of invertible formal power series with coefficients in linear differential operators, which we annotate here and throughout by the superscript $*$. We have associated projection maps

$$\mathbb{O}_k \twoheadrightarrow \mathfrak{o}_k^\rho/(\rho)^2 \cong \bigoplus_{i=1}^n \rho_i \mathfrak{o}_k \quad \text{and} \quad \mathbb{D}_k \twoheadrightarrow \mathfrak{d}_k^\rho/(\rho)^2 \cong \bigoplus_{i=1}^n \rho_i \mathfrak{d}_k. \tag{3.4}$$

Furthermore, both $\mathbb{O}_k$ and $\mathbb{D}_k$ inherit from (1.1) a Γ-action given by

$$A(\tau, \partial_\tau)\big\|_k \gamma := j_\gamma(\tau)^{-k}\, A(\gamma\tau, \partial_{\gamma\tau})\, j_\gamma^k(\tau).$$

It is these (pro-unipotent) groups with Γ-action that we will consider to define deformations of modular forms.

We make a final observation, which we apply in Section 3.4, before returning to our discussion of deformations of modular forms.

Corollary 3.1. *The isomorphism ϕ_k of Proposition 2.7 induces isomorphisms*

$$\Phi_k : \mathbb{O}_0 \cong \mathbb{O}_k \quad \textit{and} \quad \Phi_k\big|_{\mathbb{D}_0} : \mathbb{D}_0 \cong \mathbb{D}_k$$

compatible with the Γ-action.

Proof. This is an immediate consequence of the Lie group-Lie algebra correspondence, and the ρ-linear extension of the isomorphisms of Proposition 2.7. □

3.2 Formal deformations of modular forms

The setup of formal deformations parallels the one of first order deformations. In particular, we restrict to formal deformation cocycles with coefficients given by polynomials in τ. In Proposition 3.5, we examine more closely also the relation to deformations of line bundle. The key result of this section is the uniqueness stated in Theorem 3.10.

Based on the above discussion of formal deformations of line bundles and the cocycle relation in (3.1), we make the following definition.

Definition 3.2. A formal algebraic deformation cocycle of weight k is an element

$$A_\bullet \in \mathrm{Z}^1(\Gamma, \mathbb{D}_k).$$

That is, it is a map

$$A_\bullet : \Gamma \to \mathbb{D}_k$$

to (a subspace of)the space of invertible power series such that

$$A_{\gamma\delta} = A_\gamma\big\|_k \delta \cdot A_\delta \quad \text{for all } \gamma, \delta \in \Gamma.$$

Further, we say two deformation cocycles A, $\widetilde{A}$ are equivalent if they define the same cohomology class in the non-abelian group cohomology $\mathrm{H}^1(\Gamma, \mathbb{D}_k)$, which is to say there exists some $C \in \mathbb{D}_k$ such that

$$\widetilde{A}_\gamma = C\big\|_k \gamma \cdot A_\gamma \cdot C^{-1}.$$

We call the image of a formal algebraic deformation cocycle in cohomology a deformation class.

As in the case of first order deformations, we will refer to cocycles in Definition 2.4 as formal deformation cocycles, and suppress the reference to their polynomial coefficients.

The next statement asserts that deformation cocycles for different weights can be related via the isomorphism in Corollary 3.1. As such, it suffices in the remainder of this work to consider only the weight 0 theory, and we will often suppress weight from our notation where the distinction is not needed.

Lemma 3.3. *For any k, there exists an isomorphism of non-abelian cohomology sets*

$$\mathrm{H}^1(\Gamma, \mathbb{D}_0) \cong \mathrm{H}^1(\Gamma, \mathbb{D}_k)$$

induced by the isomorphism of Corollary 3.1.

Proof. This follows immediately as the isomorphism is compatible with both the group structure and the Γ-actions. It therefore maps cocycles to cocycles and coboundaries to coboundaries, given the desired isomorphism of sets. □

Remark 3.4. Recall that elements of $H^1(\Gamma, \mathfrak{o}_k)$ correspond to simultaneous deformations of the pair (Y_Γ, L_k). The projection onto $\mathcal{O}\partial_\tau$ recovers the cocycle describing just the deformation of the base space Y_Γ, while projection onto the other summand can be thought of as capturing the deformation of L_k. While these projections do not exist for formal deformations, we have an analogous construction.

A deformation of the base space $\mathcal{Y}_\Gamma$ is uniquely determined by a deformation of $\mathcal{O}_{Y_\Gamma}$, i.e. a deformation of the sheaf corresponding to weight 0 modular functions. To avoid possible confusion, we note that in Theorem 1.8 such a deformation of $\mathcal{O}_{Y_\Gamma}$ is implemented by a cocycle with values in the tangent bundle. Given a weight k deformation cocycle

$$A_\bullet \in \mathrm{Z}^1(\Gamma, \mathbb{D}_k),$$

the weight 0 cocycle $\Phi_k^{-1}(A_\bullet)$ defines a deformation of weight 0 modular forms, and hence a deformation $\mathcal{Y}_\Gamma$ of Y_Γ.

Geometrically, we may then think of the inverse isomorphism Φ_k^{-1} as an analogue of the projection $\mathcal{E}_{L_k} \to T_{Y_\Gamma}$ that appears in Theorem 1.8, allowing us to recover a deformation of the base space from the deformation of the pair.

We next determine the exact relation between deformation cocycles and deformations of the line bundle L_k.

Proposition 3.5. *Equivalent formal deformation cocycles define isomorphic deformations of L_k.*

Proof. By Lemma 3.3, it suffices to consider only the weight 0 case. Suppose $\mathcal{L}$ and $\widetilde{\mathcal{L}}$ are deformations of L_0, with corresponding deformation cocycles A_γ and $\widetilde{A}_\gamma$. Suppose A_γ and $\widetilde{A}_\gamma$ are equivalent, so that there exists $C \in \mathbb{D}_0$ such that

$$\widetilde{A}_\gamma = C\big|_k\gamma \cdot A_\gamma \cdot C^{-1}.$$

In order to show $\mathcal{L} \cong \widetilde{\mathcal{L}}$, it suffices to construct compatible isomorphisms between

$$\mathcal{L}(U) = \left\{f : \pi^{-1}(U) \to \mathbb{C}[\![\rho_1, \ldots, \rho_n]\!] : j_\gamma^{-k}(\tau) f(\gamma\tau) = A_\gamma(\tau, \partial_\tau) f(\tau)\right\}$$

and

$$\widetilde{\mathcal{L}}(U) = \left\{f : \pi^{-1}(U) \to \mathbb{C}[\![\rho_1, \ldots, \rho_n]\!] : j_\gamma^{-k}(\tau) f(\gamma\tau) = \widetilde{A}_\gamma(\tau, \partial_\tau) f(\tau)\right\}$$

for every open $U \subset \mathcal{Y}_\Gamma$. Here, $\pi : \mathbb{H} \to \mathcal{Y}_\Gamma$ is the composition

$$\mathbb{H} \to Y_\Gamma \to \mathcal{Y}_\Gamma,$$

with $\mathcal{Y}_\Gamma$ viewed as a complex analytic space.

We claim that multiplication by C defines an isomorphism for every open U. It is clearly invertible and uniform across all opens, so it just remains to check that it is well defined. Suppose $f \in \mathcal{L}(U)$. Then

$$Cf : \pi^{-1}(U) \to \mathbb{C}[\![\rho_1, \ldots, \rho_n]\!]$$

and

$$\begin{aligned} j_\gamma^{-k}(\tau) C(\gamma\tau, \partial_{\gamma\tau}) f(\gamma\tau) &= j_\gamma^{-k}(\tau) C(\gamma\tau, \partial_{\gamma\tau}) j_\gamma^k(\tau) j_\gamma^{-k}(\tau) f(\gamma\tau) \\ &= C\big|_k\gamma \cdot A_\gamma f(\tau) \\ &= C\big|_k\gamma \cdot A_\gamma \cdot C^{-1} Cf(\tau) \\ &= \widetilde{A}_\gamma \cdot Cf(\tau), \end{aligned}$$

and so $Cf \in \widetilde{\mathcal{L}}(U)$. Thus $\mathcal{L} \cong \widetilde{\mathcal{L}}$. □

We note one useful property of our deformation cocycles. For the expansion in the variables ρ_i we write

$$A_\gamma(\tau, \partial_\tau) = \sum_{m \in \mathbb{Z}^n_{\geq 0}} a_{\gamma,m}(\tau, \partial_\tau)\, \rho_1^{m_1} \cdots \rho_n^{m_n}. \tag{3.5}$$

Note that, from the definition of $\mathbb{D}_k$ as an exponential in (3.3), we must have that for all $\gamma \in \Gamma$ and all $m \in \mathbb{Z}^n$ we have $\mathrm{ord}(a_{\gamma,m}) \leq |m|$. It is also worth noting that, in weight 0, these deformation cocycles act as homomorphisms:

$$A_\gamma(\tau, \partial_\tau)(fg) = A_\gamma(\tau, \partial_\tau)(f)\, A_\gamma(\tau, \partial_\tau)(g).$$

For a power series with constant term 1, this is equivalent to $\log(A_\gamma)$ being a derivation, i.e.

$$\log(A_\gamma) \in \mathfrak{d}_0^\rho = (\rho_1, \dots, \rho_n)\, \mathfrak{d}_0[\![\rho_1, \dots, \rho_n]\!],$$

which is true, as $\mathbb{D}_0$ is defined as the exponential of $\mathfrak{d}_0^\rho$. We will later extend this to all weights to ensure that we can construct a "universal" deformation cocycle, acting as a homomorphism on the ring of modular forms.

We now wish to construct sections of the deformed line bundle. More precisely, we wish to construct a morphism of sheaves

$$\mathcal{O} \to \mathcal{O}[\![\rho_1, \dots, \rho_n]\!]$$

such that the restriction to the space of modular forms of weight k induces a map

$$\mathrm{H}^0(Y_\Gamma, L_k) \to \mathrm{H}^0(\mathcal{Y}_\Gamma, \mathcal{L}_k).$$

We will call such a map a deformation of modular forms of weight k, as in Definition 3.8 below.

Definition 3.6. A deformation section of weight k and deformation cocycle $A_\bullet$ is a trivialisation $B(\tau, \partial_\tau)$ under the coboundary map

$$\begin{aligned} \mathrm{C}^0\big(\Gamma,\, \mathbb{O}_k\big) &\to \mathrm{Z}^1\big(\Gamma,\, \mathbb{O}_k\big) \supseteq \mathrm{Z}^1\big(\Gamma,\, \mathbb{D}_k\big) \\ B &\longmapsto A_\bullet. \end{aligned}$$

That is, it is an invertible power series

$$B(\tau, \partial_\tau) \in \mathbb{O}_k \quad \text{such that } B\big\|_k \gamma \cdot B^{-1} = A_\gamma.$$

Remark 3.7. Given a deformation cocycle

$$A_\bullet \in \mathrm{Z}^1(\Gamma, \mathbb{D}_k),$$

the existence of a deformation section B of type $A_\bullet$ is equivalent to $A_\bullet$ being equivalent to the trivial deformation $A_\bullet = 1$ over $\mathbb{O}_k$. That is to say that $A_\bullet$ is a 1-coboundary in $\mathrm{Z}^1(\Gamma, \mathbb{O}_k)$. In particular, a necessary and sufficient condition for the existence of B is that the cohomology class of $A_\bullet$ vanishes under the natural map

$$\mathrm{H}^1(\Gamma, \mathbb{D}_k) \to \mathrm{H}^1(\Gamma, \mathbb{O}_k).$$

Parallelling the situation of first order deformations, a deformation section B yields a deformation of a modular form f by its action $B(\tau, \partial_\tau)\, f$ as a differential operator.

Definition 3.8. Given a deformation cocycle $A_\bullet$ of weight k and a deformation section B of type $A_\bullet$, we define a map

$$\mathcal{O} \to \mathcal{O}[\![\rho_1, \dots, \rho_n]\!]$$

on sections by $f \mapsto B(f)$. If f is a modular form of weight k, we call $B(f)$ a deformed modular form.

The following is then an immediate consequence of the definition.

Lemma 3.9. *Let $A_\bullet$ be a deformation cocycle of weight k, and B be a deformation section of type $A_\bullet$. For a modular form f of weight k, the deformed modular form transforms as*

$$\big(B(\tau, \partial_\tau)(f)\big)(\gamma\tau) = j_\gamma^k(\tau)\, A_\gamma(\tau, \partial_\tau)\, \big(B(\tau, \partial_\tau) f\big)(\tau).$$

While existence of a deformation section of a given type requires more work as explained in Remark 3.7, we can easily show that, given a deformation cocycle $A_\bullet$, there exists at most one deformation section of type $A_\bullet$.

Theorem 3.10. *Given a deformation class*

$$A_\bullet \in \ker\Big(\mathrm{H}^1(\Gamma, \mathbb{D}_k) \to \mathrm{H}^1(\Gamma, \mathbb{O}_k)\Big)$$

represented by $A_\bullet \in \mathrm{Z}^1(\Gamma, \mathbb{D}_k)$, there is at most one corresponding $\mathbb{O}_k$-coboundary whose coefficients extend to functions holomorphic at the cusp:

$$B(\tau, \partial_\tau) \in \mathbb{O}_k \text{ such that } A_\gamma = B\big\|\gamma \cdot B^{-1}.$$

Proof. Using the isomorphisms of Corollary 3.1, it suffices to establish this in weight 0. Suppose we have two power series B, $C \in \mathbb{O}_0$ such that

$$B\big\|_k\gamma \cdot B^{-1} = A_\gamma = C\big\|_k\gamma \cdot C^{-1}.$$

This implies that

$$(C^{-1} \cdot B)\big\|_k\gamma = C^{-1} \cdot B =: X$$

is an invariant function. Write

$$X(\tau) = \sum_{m\in\mathbb{Z}^n_{\geq 0}} \sum_{k=0}^{k_m} x_{m,k}(\tau)\partial_\tau^k \rho_1^{m_1} \cdots \rho_n^{m_n}$$

Considering the coefficient of the highest order derivative for each m, invariance of X implies that

$$j_\gamma^{2k_m}(\tau)x_{m,k_m}(\gamma\tau) = x_{m,k_m}(\tau)$$

and hence $x_{m,k_m}(\tau)$ is a modular form of weight $-2k_m$. There are no non-zero modular forms of negative weight, and hence $x_{m,k_m}(\tau) = 0$ unless $k_m = 0$, where $x_{m,0}(\tau)$ is a modular form of weight 0. Hence

$$X(\tau) = \sum_{m\in\mathbb{Z}^n_{\geq 0}} x_m(\tau)\rho_1^{m_1} \cdots \rho_n^{m_n}$$

with $\{x_m(\tau)\}$ a collection of weight 0 modular forms.

However, as both B and C are elements of the group $\mathbb{O}_0$, so is $X = C^{-1}B$. From the definition of $\mathbb{O}_0$ as the exponential of $\mathfrak{o}_0$, it is clear to see that there is no non-zero element x of $\mathfrak{o}_0$ such that the coefficients of $\exp(x)$ are differential operators of degree 0. In particular, there is no non-zero x such that $X = \exp(x)$. Thus, we must have $X = 1$. □

3.3 Extension of first order deformations

The goal of this section is to complement Theorem 3.10 and show that first order deformation cocycles extend to formal ones.

Remark 3.11. It is as yet unclear whether Bogo's construction yields an algebraic prolongation of our common first order deformation. By direct computation, one may show that Bogo's construction agrees with ours up to second order (cf. Section 3.4). The recursive nature of Bogo's construction would suggest his deformation is algebraic in the sense of this section, but it is as yet unclear how to prove this.

Let us first consider the first order approximations

$$A_\gamma(\tau, \partial_\tau) = 1 + \sum_{i=1}^n a_{\gamma,i}(\tau, \partial_\tau)\rho_i + O\big(\rho^2\big),$$

where we use $O(\rho^2)$ to refer to any terms of order at least 2 in $\rho_1, \ldots, \rho_n$.

The non-abelian cocycle condition

$$A_{\gamma\delta} = A_\gamma\big\|_k\delta \cdot A_\delta$$

reduces to the linear cocycle condition

$$a_{\gamma\delta,i} = a_{\gamma,i}\big\|_k\delta + a_{\delta,i} \quad \text{for all } i \text{ such that } 1 \leq i \leq n.$$

From the results of Section 2, specifically Proposition 2.9 we know that solutions to this cocycle condition are constructed from weight 4 modular forms.

Given an algebraic deformation to first order, the natural question to ask is whether it can be prolonged to a formal deformation. As Γ has virtual cohomological dimension 2, this is always possible.

Theorem 3.12. *The natural projection map*

$$\mathrm{Z}^1(\Gamma, \mathbb{D}_k) \to \bigoplus_{i=1}^n \mathrm{Z}^1(\Gamma, \mathfrak{d}_k)$$

associated with the one in (3.4) *is surjective. That is, given* $a_{\bullet,i} \in \mathrm{Z}^1(\Gamma, \mathfrak{d}_k)$ *for* $i = 1, \dots, n$*, it is always possible to find* $A_\bullet \in \mathrm{Z}^1(\Gamma, \mathbb{D}_k)$ *such that*

$$A_\bullet = 1 + \sum_{i=1}^{n} a_{\bullet,i}\rho_i + O\big(\rho^2\big).$$

In other words, given a first order approximation of a deformation cocycle of weight k*, it is always possible to prolong it to a formal deformation cocycle.*

Proof. Denote by $\mathbb{D}_k/(\rho)^N$ the image of the natural map

$$\mathbb{D}_k \to \mathbb{C}[\tau, \partial_\tau][\![\rho_1, \dots, \rho_n]\!]/(\rho)^N,$$

and note that every element of $\mathbb{D}_k$ is uniquely determined by its image in the family $\big\{\mathbb{D}_k/(\rho)^N\big\}_{N\geq 1}$. That is to say that $\mathbb{D}_k$ is the projective limit of the system $\big\{\mathbb{D}_k/(\rho)^N\big\}_{N\geq 1}$ with the obvious surjections. As such, it suffices to show that the natural projection

$$\mathrm{Z}^1\big(\Gamma,\, \mathbb{D}_k/(\rho)^{N+1}\big) \to \mathrm{Z}^1\big(\Gamma,\, \mathbb{D}_k/(\rho)^N\big)$$

is surjective for every $N \geq 1$. We first show the corresponding fact for the cohomology groups. For every $N \geq 1$, we have a short exact sequence of Γ-groups

$$0 \to (\rho)^N \mathfrak{d}_k \to \mathbb{D}_k/(\rho)^{N+1} \to \mathbb{D}_k/(\rho)^N \to 0.$$

As $\mathfrak{d}_k$ is an abelian group in this context, this implies that we have an exact sequence of pointed sets

$$\begin{aligned} 0 \to (\rho)^N \big(\mathfrak{d}_k\big)^\Gamma \to \big(\mathbb{D}_k/(\rho)^{N+1}\big)^\Gamma \to \big(\mathbb{D}_k/(\rho)^N\big)^\Gamma \to \\ \to \mathrm{H}^1\big(\Gamma, \mathfrak{d}_k\big)^{\oplus n} \to \mathrm{H}^1\big(\Gamma, \mathbb{D}_k/(\rho)^{N+1}\big) \to \mathrm{H}^1\big(\Gamma, \mathbb{D}_k/(\rho)^N\big) \to \mathrm{H}^2\big(\Gamma, \mathfrak{d}_k\big)^{\oplus n}. \end{aligned}$$

As $\mathrm{H}^2(\Gamma, V) = 0$ for any finite dimensional representation V, the final term in this sequence vanishes, and hence

$$\mathrm{H}^1\big(\Gamma, \mathbb{D}_k/(\rho)^{N+1}\big) \to \mathrm{H}^1\big(\Gamma, \mathbb{D}_k/(\rho)^N\big)$$

is surjective. As such, for any element $A_\bullet^{(N)} \in \mathrm{Z}^1\big(\Gamma, \mathbb{D}_k/(\rho)^N\big)$, there exists

$$A_\bullet^{(N+1)} \in \mathrm{Z}^1\big(\Gamma, \mathbb{D}_k/(\rho)^{N+1}\big) \text{ and } C_{(N)} \in \mathbb{D}_k/(\rho)^N$$

such that

$$A_\gamma^{(N+1)} \equiv C_{(N)} \big\|_k \gamma A_\gamma^{(N)} C_{(N)}^{-1} \pmod{(\rho)^N}.$$

Taking any lift C of $C_{(N)}$ in $\mathbb{D}_k$, the cocycle $C^{-1}\big\|_k \bullet A_\bullet^{(N+1)} C$ will be a lift of $A_\bullet^{(N)}$ in $\mathbb{D}/(\rho)^{N+1}$, giving surjectivity at the level of cocycles. □

3.4 Calculation of second order deformations

We have shown that we can extend any first order deformation cocycle to a formal algebraic deformation cocycle. However, the arguments of Proposition 3.12 are not constructive, and so we will here illustrate a method for explicit computation. Specifically, we will compute an example of a deformation cocycle to second order.

In order to do this, we consider the Lie algebraic perspective on our cocycle conditions. We do so using (a special case of) the Baker-Campbell-Hausdorff formula [15, 6].

Theorem 3.13. *Let* $X, Y \in \mathfrak{g}$ *be elements of a pro-nilpotent Lie algebra. Then there exists* $Z \in \mathfrak{g}$ *such that* $\exp(X)\exp(Y) = \exp(Z)$*, given by the Baker-Campbell-Hausdorff (BCH) formula*

$$Z = \mathrm{BCH}(X, Y) = X + Y + \tfrac{1}{2}[X, Y] + \cdots \tag{3.6}$$

Remark 3.14. Convergence of the Baker-Campbell-Hausdorff formula is not assured for general Lie algebras and Lie groups. However, if one has some meaningful notion of convergent series of elements of the Lie algebra, the above proposition holds for sufficiently "small" X and Y.

We will be considering $\mathfrak{o}_k^\rho$, as defined in Section 2, and its Lie subalgebra $\mathfrak{d}_k^\rho$. These are pro-nilpotent Lie algebras, and the coefficient of each $\rho_1^{m_1}\dots\rho_n^{m_n}$ in the BCH formula will only involve finitely many terms. As such, we may freely apply the BCH formula to our setting.

Suppose we have a deformation cocycle $A_\bullet = \exp(a_\bullet)$ of weight k for some $a_\bullet \in \mathfrak{d}_k^\rho$. Then the cocycle condition is equivalent to

$$a_{\gamma\delta} = \mathrm{BCH}\big(a_\gamma\big\|_k\delta, a_\delta\big). \tag{3.7}$$

The existence of a deformation section is then equivalent to the existence of a $b \in \mathfrak{o}_k^\rho$ such that

$$a_\gamma = \mathrm{BCH}(b\big\|_k\gamma, -b). \tag{3.8}$$

By considering the coefficients of each $\rho_1^{m_1}\dots\rho_n^{m_n}$ in Equations (3.7) and (3.8), we obtain a system of inhomogeneous cocycle equations, which we can solve recursively.

Definition 3.15. An element $a_\bullet \in \mathfrak{d}_k^\rho$ satisfying Equation (3.7) is called a Lie cocycle of weight k. Given a Lie cocycle $a_\bullet$, an element $b \in \mathfrak{o}_k^\rho$ satisfying Equation (3.8) is called a Lie deformation section of type $a_\bullet$.

4 Universal and canonical deformations

4.2 Existence of deformation sections

The goal of this section is to establish the existence of formal deformation sections associated to any given deformation cocycle. We achieve this in Theorem 4.3. However, much like the proof of Theorem 3.12, the proof of this is entirely non constructive. As such, we present a more explicit recipe in Proposition 4.5 and Corollary 4.7 for constructing a Lie section, modulo some subtleties surrounding convergence of integrals.

Theorem 4.3. *Given a deformation cocycle $A_\bullet$ of weight k, there exists a deformation section B of weight k and type $A_\bullet$.*

Proof. As usual, we only need to establish the result in weight 0. This is equivalent to showing that the natural map

$$\mathrm{H}^1(\Gamma, \mathbb{D}_k) \to \mathrm{H}^1(\Gamma, \mathbb{O}_k)$$

is the zero map. Analogously to the proof of Theorem 3.12, we have an exact sequence of pointed sets

$$\cdots \to \mathrm{H}^1\big(\Gamma, \mathfrak{o}_0\big)^{\oplus n} \to \mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^{N+1}\big) \to \mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^N\big) \to 0,$$

where we denote by $\mathbb{O}_0/(\rho)^N$ the image of the natural map

$$\mathbb{O}_0 \to \mathcal{O}[\partial_\tau][\![\rho_1, \dots, \rho_n]\!]/(\rho)^N.$$

By Lemma 2.8, $H^1(\Gamma, \mathfrak{o}_k) = 0$. Hence, we have an isomorphism

$$\mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^{N+1}\big) \cong \mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^N\big)$$

for every $N \geq 0$. But

$$\mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^1\big) \cong \bigoplus_{i=1}^n \mathrm{H}^1\big(\Gamma, \mathfrak{o}_0\big) = 0,$$

and so $\mathrm{H}^1\big(\Gamma, \mathbb{O}_0/(\rho)^N\big) = 0$ for every $N \geq 0$. Taking the projective limit, we see that

$$\mathrm{H}^1\big(\Gamma, \mathbb{O}_0\big) = 0,$$

giving the desired result. □

Remark 4.4. The vanishing of $\mathrm{H}^1(\Gamma, \mathbb{O}_0)$ implies that the connecting map

$$\mathrm{H}^0(\Gamma, \mathbb{O}_0/\mathbb{D}_0) \to \mathrm{H}^1(\Gamma, \mathbb{D}_0)$$

in the exact sequence associated to

$$0 \to \mathbb{D}_0 \to \mathbb{O}_0 \to \mathbb{O}_0/\mathbb{D}_0 \to 0$$

is surjective, and furthermore that a coboundary realising $A_\bullet$ is unique up to an element of $\mathbb{D}_0$. One might hope that if we could extend Theorem 4.3 to give deformation sections with "nice" poles at the cusp, we could always ensure a deformation section that was holomorphic at the cusp.

Indeed, if we were to replace $\mathfrak{o}_0$ with the Γ-submodule $\mathfrak{o}_0^{\leq 2}$ consisting of differential operators whose coefficients have poles of order at most 2, then an analogue of Lemma 2.8 and the long exact sequence associated to

$$0 \to \mathfrak{d}_0 \to \mathfrak{o}_0^{\leq 2} \to \mathfrak{o}_0^{\leq 2}/\mathfrak{d}_0 \to 0$$

would imply the existence of a deformation section extending holomorphically to infinity. Given the triple integral appearing in the constructive argument of Proposition 4.5, we believe it should be possible to always construct a coboundary B realising a deformation cocycle $A_\bullet$ that has coefficients holomorphic at the cusp.

As mentioned previously, this approach is entirely non-constructive. We offer the following pair of results demonstrating a method for explicitly constructing a Lie section, assuming only that we can find a coboundary realisation for certain linear cocycles. Since we will consider deformations in several parameters, it is convenient to write $|m| = m_1 + \cdots + m_n$ for n-tuples m of numbers.

Proposition 4.5. *Given a pair of a* first order *algebraic deformation and first order deformation section $(a_\bullet, b)$ of weight k, it is possible to find a pair of a* formal *deformation cycle and deformation section $(A_\bullet, B)$ such that*

$$A_\bullet \equiv \exp(a_\bullet) \equiv 1 + a_\bullet \quad \textit{and} \quad B \equiv \exp(b) \equiv 1 + b \qquad (\mathrm{mod}\,(\rho)^2).$$

Proof. We consider only the weight 0 case, as the other cases follow by Corollary 3.1. Working at the level of Lie algebras we use the ansatz

$$(A_\bullet,\, B) = \big(\exp(a_\bullet^\Sigma),\, \exp(b^\Sigma)\big)$$

with

$$a_\bullet^\Sigma = \sum a_\bullet^{(m)}\, \rho_1^{m_1} \cdots \rho_n^{m_n} \in \mathfrak{d}_0^\rho, \quad b^\Sigma = \sum b^{(m)}\, \rho_1^{m_1} \cdots \rho_n^{m_n} \in \mathfrak{o}_0^\rho,$$

where $m = (m_1, \ldots, m_n)$ runs through n-tuples of nonnegative integers, and

$$a_\bullet^\Sigma \equiv a_\bullet \quad \text{and} \quad b^\Sigma \equiv b \qquad (\mathrm{mod}\,(\rho)^2).$$

We need to ensure that

$$a^\Sigma_{\gamma\delta} = \mathrm{BCH}\big(a^\Sigma_\gamma\big\|_0\delta,\, a^\Sigma_\delta\big) \quad \text{and} \quad a^\Sigma_\gamma = \mathrm{BCH}\big(b^\Sigma\big\|_0\gamma,\, -b^\Sigma\big),$$

In fact, it suffices to find b^Σ such that

$$a^\Sigma_\gamma := \mathrm{BCH}\big(b^\Sigma\|_0\gamma,\, -b^\Sigma\big) \tag{4.2}$$

defines an element of $\mathfrak{d}_0^\rho$.

As such, suppose we have constructed the section b^Σ to order N and consider an index $m = (m_1, \ldots, m_n)$ with $|m| = N+1$. Consider the coefficient of $\rho_1^{m_1} \ldots \rho_n^{m_n}$ of Equation (4.2). In order for a^Σ_γ to be an element of $\mathfrak{d}_0^\rho$, we require that the left hand side be a quadratic polynomial. Hence, it suffices to find $b^{(m)}$ such that the coefficient of $\rho_1^{m_1} \ldots \rho_n^{m_n}$ of the right hand side:

$$j_\gamma^2(\tau)\, b^{(m)}(\gamma\tau) - b^{(m)}(\tau) + \mathrm{BCH}\big(b^\Sigma;\gamma\big)^{(m)}(\tau) \tag{4.3}$$

is a quadratic polynomial. Here we use $\mathrm{BCH}(b^\Sigma;\gamma)^{(m)}(\tau)$ to denote the coefficient of the monomial $\rho_1^{m_1} \cdots \rho_n^{m_n}$ in the non-linear part of the BCH-formula (3.6)

$$\tfrac{1}{2}\big[b^\Sigma\big\|_0\gamma,\, -b^\Sigma\big] + \tfrac{1}{12}\Big[b^\Sigma\big\|_0\gamma,\, \big[b^\Sigma\big\|_0\gamma,\, -b^\Sigma\big]\Big] + \cdots.$$

Notably, $\mathrm{BCH}(b^\Sigma;\gamma)^{(m)}$ is defined in terms of lower order terms of b^Σ, which we assume to have constructed.

As we want the expression (4.3) to be a quadratic polynomial, it must be annihilated by taking the third derivative. We therefore wish to find $b^{(m)}$ such that

$$j_\gamma^{-4}(\tau)\frac{\mathrm{d}^3 b^{(m)}}{\mathrm{d}\tau^3}(\gamma\tau) - \frac{\mathrm{d}^3 b^{(m)}}{\mathrm{d}\tau^3}(\tau) = -\frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}\big(b^\Sigma;\gamma\big)^{(m)}(\tau), \tag{4.4}$$

where we use Bol's identity to simplify the calculation of the first term.

By Lemma 4.6, the third derivative of $\mathrm{BCH}(b^\Sigma;\gamma)^\gamma$ satisfies a cocycle condition

$$\frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}\big(b^\Sigma;\gamma\delta\big)^{(m)}(\tau) - j_\delta^{-4}(\tau)\frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}\big(b^\Sigma;\gamma\big)^{(m)}(\delta\tau) - \frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}\big(b^\Sigma;\delta\big)^{(m)}(\tau) = 0.$$

We can therefore view Equation (4.4) as an equation in group cohomology, realising the cocycle $({}^{\mathrm{d}^3}\!/_{\mathrm{d}\tau^3})\,\mathrm{BCH}(b^\Sigma;\gamma\delta)^{(m)}$ as a coboundary. Hence, the above equation has a solution if and only if $({}^{\mathrm{d}^3}\!/_{\mathrm{d}\tau^3})\,\mathrm{BCH}(b^\Sigma;\gamma\delta)^{(m)}$ defines the zero element in the (abelian) group cohomology $\mathrm{H}^1(\Gamma,\mathcal{O})$, where we view $\mathcal{O}$ as a Γ-module.

As previously, we may identify group cohomology with sheaf cohomology of the modular curve to conclude that

$$\mathrm{H}^1(\Gamma,\mathcal{O}) \cong \mathrm{H}^1(Y_\Gamma,\Omega^2) = 0,$$

where Ω^2 is the symmetric square of Kähler-differentials on Y_Γ. Thus, we can always realise $({}^{\mathrm{d}^3}\!/_{\mathrm{d}\tau^3})\,\mathrm{BCH}(b^\Sigma;\gamma\delta)^{(m)}$ as a coboundary, and hence find $({}^{\mathrm{d}^3}\!/_{\mathrm{d}\tau^3})b^{(m)}$. Any convergent triple integral of which can then be taken as $b^{(m)}$, and the existence of such a choice of coboundary is guaranteed by Theorem 4.3. □

In the next lemma, to simplify notation, we write b instead of b^Σ from the proof of Proposition 4.5.

Lemma 4.6. *Let $b \in \mathfrak{o}_0^\rho$ be a finite series containing only terms of total degree at most N, such that $\mathrm{BCH}(b;\gamma)^{(m)} \in \mathfrak{d}_0^\rho$ for all $m = (m_1,\dots,m_n)$ with $|m| \leq N$ and all $\gamma \in \Gamma$. Then for every $m = (m_1,\dots,m_n)$ with $|m| = N+1$, the third derivative of $\mathrm{BCH}(b;\gamma)^{(m)}$ satisfies the cocycle condition*

$$\frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}(b;\gamma\delta)^{(m)}(\tau) - j_\delta^{-4}(\tau)\frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}(b;\gamma)^{(m)}(\delta\tau) - \frac{\mathrm{d}^3}{\mathrm{d}\tau^3}\,\mathrm{BCH}(b;\delta)^{(m)}(\tau) = 0.$$

Proof. Define $c_\gamma := \mathrm{BCH}(b\|_0\gamma, -b)$. By definition $c_\bullet$ is a Lie coboundary and hence a Lie cocycle. In particular

$$c_{\gamma\delta} = \mathrm{BCH}\big(c_\gamma\big\|_0\delta, c_\delta\big)$$

for all γ, $\delta \in \Gamma$. Fix a multi-index $m = (m_1,\dots,m_n)$ such that $|m| = N+1$. Expanding out the cocycle equation, we have that

$$c_{\gamma\delta}^{(m)}(\tau) = j_\delta^2(\tau)c_\gamma^{(m)}(\delta\tau) + c_\delta^{(m)}(\tau) + \mathrm{BCH}\big(c_\gamma\big\|_0\delta, c_\delta\big)^{(m)}.$$

The third term is a Lie polynomial involving only terms of $c_\bullet$ of order at most N, which by assumption are all elements of $\mathfrak{d}_0^\rho$. As $\mathfrak{d}_0^\rho$ is closed as a Lie algebra, this implies that

$$\mathrm{BCH}(c_\gamma\big\|_0\delta, c_\delta)^{(m)}\partial_\tau \in \mathfrak{d}_0^\rho$$

and, in particular $\mathrm{BCH}(c_\gamma\big\|_0\delta, c_\delta)^{(m)}$ is a quadratic polynomial in τ. Hence, the non-linear part with respect to ∂_τ of the above equation is annihilated by the third derivative. Employing Bol's identity to simplify calculations, we have that

$$\frac{\mathrm{d}^3 c_{\gamma\delta}^{(m)}}{\mathrm{d}\tau^3}(\tau) = j_\delta^{-4}(\tau)\frac{\mathrm{d}^3 c_\gamma^{(m)}}{\mathrm{d}\tau^3}(\delta\tau) + \frac{\mathrm{d}^3 c_\delta^{(m)}}{\mathrm{d}\tau^3}(\tau)$$

from which the claim follows. □

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